Week 6: Attention Mechanisms and Self-Attention

Redesigned Question Bank — Set 1 (Questions 1–10)

SIS 3204Y(5) Natural Language Processing

Assessment design note: Numerical questions use only arithmetic explicitly supported by the Week 6 lecture: simple dot products or weighted sums. Softmax values are not required to be calculated manually.

Question 1 — Why Do We Need Attention? [6 marks]

Question: Explain the encoder-decoder bottleneck and how attention helps to overcome it.

Sample Answer

Without attention, a traditional encoder-decoder model attempts to represent the whole input sequence using one fixed context representation. Important information can be difficult to preserve, especially when the sequence is long.

Attention allows the decoder to look back at the encoder hidden states and give more weight to the input positions that are most relevant for the current output. The context used by the decoder can therefore change from one decoding step to another.

Mark Breakdown

Criterion	Marks
Fixed-context bottleneck explained	2
Difficulty with long sequences identified	1
Decoder can access encoder hidden states	1
Relevant positions receive higher attention	1
Context can change across decoding steps	1
Total	**6**

Question 2 — Attention Scores, Weights and Context Vector [8 marks]

Question: Explain the difference between an attention score, an attention weight and a context vector.

Sample Answer

An attention score measures how relevant an input representation is to the current query or decoder state. The scores are passed through softmax to obtain attention weights. These weights are positive and sum to 1.

The context vector is obtained by taking a weighted sum of the relevant hidden-state or Value vectors. It therefore combines information from the input according to the learned attention weights.

Mark Breakdown

Criterion	Marks
Attention score correctly explained	2
Softmax conversion to weights	2
Properties of weights stated	1
Context vector described as weighted sum	2
Relationship among all three concepts is clear	1
Total	**8**

Question 3 — Interpreting Attention in Translation [8 marks]

Question: Consider the translation “The cat sat on the mat.” → “Le chat était assis sur le tapis.” Explain how attention may help when the decoder generates different French words.

Sample Answer

At each output step, the decoder can assign different attention weights to the English input words. When generating “chat”, information associated with “cat” may be particularly relevant. When generating “était” or “assis”, information associated with “sat” may become more relevant.

The attention weights are used to build a context vector for that decoding step. Therefore, the model is not forced to use exactly the same source information for every output word.

Mark Breakdown

Criterion	Marks
Attention can vary at each decoding step	2
Correct use of translation example	2
Relevant source positions can receive higher weight	2
Context vector supports current output generation	2
Total	**8**

Question 4 — What is Self-Attention? [8 marks]

Question: Define self-attention. How is it different from encoder-decoder attention?

Sample Answer

Self-attention allows each token to consider other tokens in the same sequence when producing its contextual representation.

In encoder-decoder attention, a decoder query attends to representations from the encoder. In self-attention, the Query, Key and Value representations are derived from the same sequence. This enables each token representation to incorporate information from other relevant tokens in that sequence.

Mark Breakdown

Criterion	Marks
Self-attention correctly defined	2
Same-sequence relationship explained	2
Encoder-decoder attention correctly contrasted	2
Q, K and V from same sequence stated	1
Contextual representation explained	1
Total	**8**

Question 5 — Query, Key and Value [9 marks]

Question: Explain the roles of Query (Q), Key (K) and Value (V) in self-attention. State how they are obtained from the input representation X.

Sample Answer

The Query represents what a token is looking for. The Key provides information used to determine how relevant another token is to that Query. The Value contains the information that will be combined to form the contextualised output.

They are produced using learned linear transformations:
Q = XW_Q
K = XW_K
V = XW_V

Queries are compared with Keys to obtain relevance scores, and the resulting attention weights are used to combine the Values.

Mark Breakdown

Criterion	Marks
Query role	2
Key role	2
Value role	2
Q = XW_Q	1
K = XW_K	1
V = XW_V	1
Total	**9**

Question 6 — Scaled Dot-Product Attention [10 marks]

Question: Write the scaled dot-product attention formula and explain the purpose of each part of the formula.

Sample Answer

The formula is:

Attention(Q,K,V) = softmax((QK^T)/√d_k)V

QK^T calculates relevance or similarity scores between Queries and Keys. Dividing by √d_k keeps the scores from becoming excessively large when the Key dimension is large. Softmax converts the scaled scores into attention weights. Finally, the attention weights are applied to V to produce contextualised output representations.

Mark Breakdown

Criterion	Marks
Correct formula	2
QK^T explained	2
Scaling by √d_k explained	2
Softmax explained	2
Application to V/output explained	2
Total	**10**

Question 7 — Simple Attention Calculation [8 marks]

Question: For one token, the attention weights are [0.2, 0.5, 0.3]. The corresponding Value vectors are V1=[1,0], V2=[0,2] and V3=[1,1].

Calculate the contextual output vector. You do NOT need to calculate softmax.

Sample Answer

Output = 0.2V1 + 0.5V2 + 0.3V3

= 0.2[1,0] + 0.5[0,2] + 0.3[1,1]

= [0.2,0] + [0,1.0] + [0.3,0.3]

= [0.5, 1.3]

The output combines information from all three Value vectors, with V2 having the largest contribution because it has the largest attention weight.

Mark Breakdown

Criterion	Marks
Correct weighted-sum expression	2
Correct contribution from V1	1
Correct contribution from V2	1
Correct contribution from V3	1
Correct final vector [0.5, 1.3]	2
Interpretation of largest contribution	1
Total	**8**

Question 8 — What Can Self-Attention Represent? [8 marks]

Question: Give four types of relationships that self-attention can represent within a sentence. Briefly explain each one.

Sample Answer

Self-attention can represent several kinds of contextual relationships. These include local syntactic relationships, long-range dependencies, referential relationships such as links involving pronouns, and semantic/contextual relationships between words.

For example, a verb may relate strongly to its subject, a word may depend on another word much earlier in a sentence, a pronoun may draw information from a relevant noun, and the meaning of a word such as “bank” may be clarified by surrounding words such as “loan”.

Mark Breakdown

Criterion	Marks
Local/syntactic relationship	2
Long-range dependency	2
Referential/contextual relationship	2
Semantic/context-sensitive relationship	2
Total	**8**

Question 9 — Multi-Head Attention [10 marks]

Question: What is multi-head attention and why can it be more useful than using only one attention head?

Sample Answer

Multi-head attention performs several attention operations in parallel using different learned projections of Q, K and V. Each head produces an attention output. The head outputs are then concatenated and passed through an output projection.

Using several heads gives the model the capacity to learn multiple attention patterns and representation subspaces at the same layer. Different heads may focus on different relationships, although a head should not automatically be assumed to correspond to one fixed linguistic rule.

Mark Breakdown

Criterion	Marks
Several attention heads operate in parallel	2
Different learned projections of Q/K/V	2
Each head produces an output	1
Outputs concatenated/projected	2
Benefit of multiple patterns/subspaces	2
Appropriate caution about interpreting heads	1
Total	**10**

Question 10 — From Self-Attention to Transformers [10 marks]

Question: Explain how self-attention leads towards the Transformer architecture. Identify the main ideas introduced in Week 6 that form this bridge.

Sample Answer

Self-attention produces contextual representations by allowing tokens to attend to other tokens in the same sequence. Scaled dot-product attention defines the core computation, while multi-head attention performs several such attention operations in parallel.

To move towards Transformers, token representations also need positional information because self-attention itself does not process tokens recurrently. Multi-head self-attention and feed-forward processing are then organised into Transformer blocks. The complete Transformer architecture is studied separately.

Mark Breakdown

Criterion	Marks
Self-attention contextualisation	2
Scaled dot-product attention	1
Multi-head attention	2
Positional information	2
Feed-forward processing	1
Transformer blocks/bridge correctly described	2
Total	**10**